
Arthritis Progression Prediction Using NLP

Neelam More [20038484], nmore2@stevens.edu
Viyasha Thakkar [20049574], ythakkar@stevens.edu

Stevens Institution of Technology
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1 Introduction

Arthritis is a leading cause of disability, affecting millions worldwide. Early prediction of disease progression can help medical professionals intervene at the right time. While traditional methods rely on medical imaging and lab tests, clinical notes and patient histories contain valuable unstructured data that can be analysed using NLP techniques. This project focuses on leveraging machine learning models trained on arthritis-related text data to predict disease progression. The objective is to provide an AI-driven solution that can assist doctors in making informed decisions regarding patient care.

2 Problem Formulation

This project is a **classification task** where we predict the progression of arthritis in patients based on textual clinical notes. The goal is to train an NLP-based model to classify arthritis severity into different stages (mild, moderate, severe) or predict the likelihood of disease worsening. Formally, we define our task as:

f: $X \rightarrow Y$

Where X represents the set of patient records (clinical notes, diagnostic history), and Y is the classification label indicating disease progression. Our objective is to minimize classification error and improve early detection accuracy.

3 Methods

We will approach the problem by performing the following steps:

3.1 Data Collection

- Obtain arthritis-related datasets containing clinical notes, patient reports, and diagnostic histories.

3.2 Text Preprocessing

- Clean and preprocess textual data, including tokenization, stemming, and stop-word removal.

3.3 Feature Engineering

- Extract key medical terms, disease symptoms, and progression indicators using Named Entity Recognition(NER).

3.4 Model Development

- Train deep learning models such as BERT, RoBERTa, and LSTMs for predictive analysis.

3.5 Evaluation

- Measure model performance using classification metrics.

4 Dataset and Experiments

4.1 Dataset sources

- **Osteoarthritis Initiative(OAI):** clinical records and diagnostic reports [arXiv:1810.04805].
- **Emory Knee Radiograph Dataset:** clinical notes and imaging data [arXiv:2411.00866].
- Additional arthritis-related clinical datasets with progression labels [BMJ, 2024].

4.2 Experimental Setup

- **Hardware/Software:** Python, PyTorch, HuggingFace Transformers, Scikit-learn.
- **Training:** 50–100 epochs depending on model, batch sizes 16–32, learning rate $1e-5$ to $5 \times 1e-5$.
- **Comparison models:** Logistic Regression, Random Forest, TF-IDF, and Word2Vec-based classifiers.

4.3 Evaluation Metrics

- **Accuracy:** overall prediction correctness.
- **Precision, Recall, F1-score:** to handle class imbalance.
- **AUC-ROC:** classification performance on different severity levels.

4.4 Planned Results

- Performance tables comparing BERT, RoBERTa, LSTMs vs baselines.
- Feature importance analysis to highlight predictive clinical terms.
- Error analysis on misclassified cases.

5 Project Management

Team Members and Responsibilities

- **Viyasha Thakkar [20049574]**– Data Collection , Preprocessing & Evaluation
- **Neelam More [20038484]**– Model Training, Fine-tuning & Experimentation

Timeline

- **Week 1-2:** Literature review and dataset collection
- **Week 3-4:** Data preprocessing and exploratory data analysis
- **Week 5-6:** Model development (BERT, RoBERTa, LSTMs)
- **Week 7:** Final evaluation and comparison of models
- **Week 8:** Report writing and presentation preparation

6 Conclusion

We will summarize our key contributions and suggest directions for further research:

- **Contributions:** Developing an NLP-based predictive model for arthritis progression.
- **Limitations:** Challenges in handling medical text complexity and dataset limitations.
- **Future Work:** Potential for integrating multimodal data (e.g., combining text with medical imaging).

References

Datasets:

[1]. Osteoarthritis Initiative (OAI): A publicly available dataset containing clinical records, diagnostic reports, and radiographic imaging related to osteoarthritis.

[Access here- https://arxiv.org/abs/1810.04805](https://arxiv.org/abs/1810.04805)

[2]. Emory Knee Radiograph Dataset: A large dataset of knee radiographs with clinical information. [arXiv Paper- https://arxiv.org/abs/2411.00866](https://arxiv.org/abs/2411.00866)

[3]. Predicting rapid progression in knee osteoarthritis: a novel and interpretable automated machine learning approach, with specific focus on young patients and early disease: <https://nda.nih.gov/oai/>.